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Module 5: Communication, Homeostasis and Energy

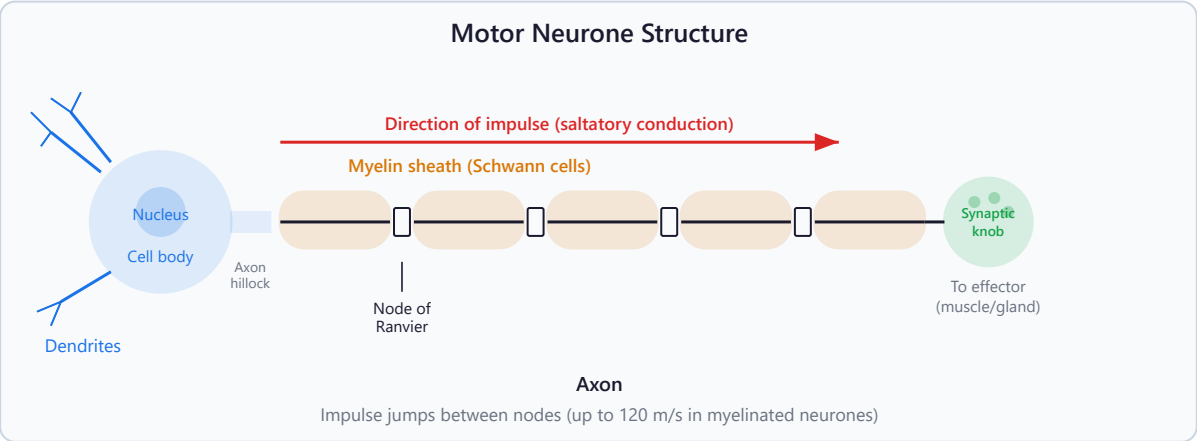
OCR A A-Level Biology (H420)

5.1 Neuronal Communication

The Nervous System

The nervous system provides rapid, short-lived, targeted communication using electrical impulses transmitted along neurones. It consists of the central nervous system (CNS: brain and spinal cord) and the peripheral nervous system (PNS: sensory and motor neurones). The PNS is subdivided into the somatic nervous system (voluntary, controls skeletal muscle) and the autonomic nervous system (involuntary, controls smooth muscle, cardiac muscle, and glands).

A reflex arc provides rapid, involuntary responses to stimuli: stimulus is detected by a receptor, sensory neurone transmits impulse to the CNS, relay neurone in the spinal cord connects to a motor neurone, which transmits impulse to an effector (muscle or gland) that produces the response. Reflexes are fast because they involve few synapses, are involuntary (no conscious processing delays), and follow predetermined neural pathways.



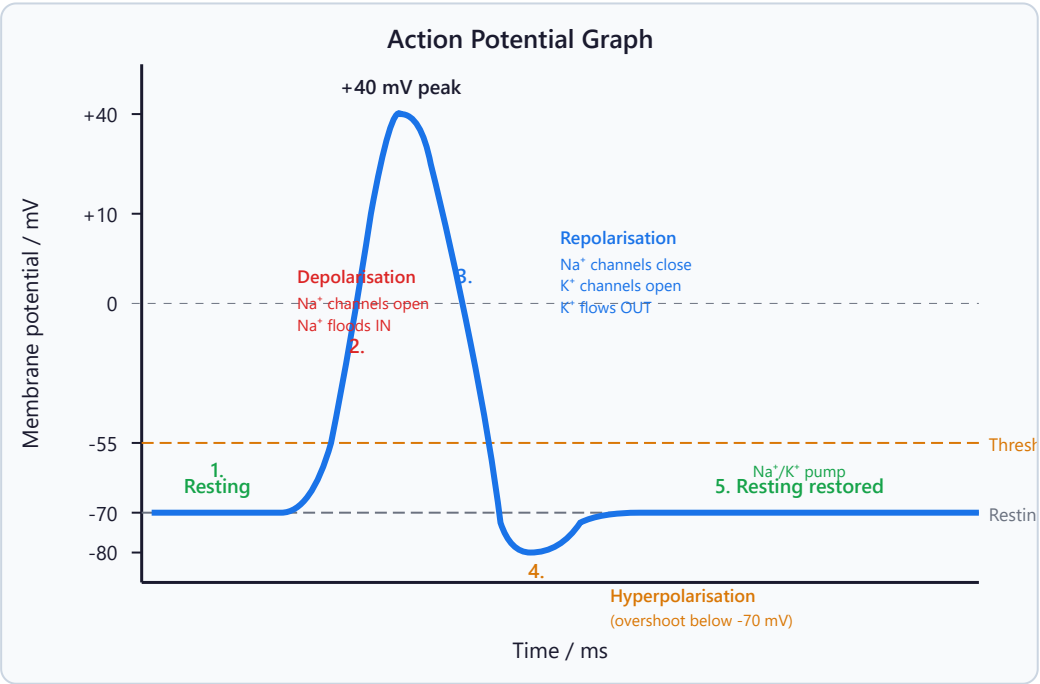
The Resting Potential

At rest, a neurone has a potential difference of approximately -70 mV across its membrane (inside negative relative to outside). This resting potential is maintained by:

- The sodium-potassium pump (Na+/K+ ATPase) actively transporting 3 Na+ ions out and 2 K+ ions in per cycle, using ATP
- Greater permeability of the membrane to K+ than Na+ (more K+ leak channels are open at rest), allowing K+ to diffuse out down its concentration gradient
- Large organic anions trapped inside the cell contributing to the negative interior

The Action Potential

When a stimulus depolarises the membrane to the threshold potential (approximately -55 mV), voltage-gated sodium channels open rapidly, Na+ floods in down its electrochemical gradient, and the membrane depolarises to approximately +40 mV. This is followed by repolarisation as sodium channels inactivate and voltage-gated potassium channels open, K+ flows out, and the membrane returns to negative. A brief hyperpolarisation (more negative than -70 mV) occurs before the resting potential is restored.



All-or-Nothing Principle: If the threshold is reached, a full action potential of constant amplitude is generated regardless of stimulus strength. If threshold is not reached, no action

potential occurs. Stimulus intensity is coded by the frequency of action potentials (stronger stimuli produce more frequent impulses) rather than by amplitude.

Impulse Transmission

In myelinated neurones, Schwann cells wrap around the axon forming a myelin sheath interrupted by nodes of Ranvier (gaps between adjacent Schwann cells). The myelin sheath acts as an electrical insulator, preventing ion flow across the membrane. Depolarisation can only occur at the nodes, so the impulse jumps from node to node (saltatory conduction), greatly increasing transmission speed (up to 120 m/s compared to 1 m/s in unmyelinated neurones).

The refractory period (the brief time after an action potential during which the membrane cannot be restimulated) ensures unidirectional propagation of the impulse and limits the maximum frequency of impulses. The absolute refractory period (sodium channels inactivated, no stimulus can trigger another action potential) is followed by the relative refractory period (some channels have reset, but a stronger-than-normal stimulus is needed).

Synaptic Transmission

Synapses are junctions between neurones where signals are transmitted chemically across the synaptic cleft (approximately 20 nm gap). The sequence of events at a cholinergic synapse is:

1. Action potential arrives at the presynaptic terminal (synaptic knob)
2. Voltage-gated calcium channels open; Ca2+ ions flow into the presynaptic terminal
3. Calcium ions cause synaptic vesicles containing acetylcholine (ACh) to fuse with the presynaptic membrane (exocytosis)
4. ACh diffuses across the synaptic cleft and binds to specific receptors on the postsynaptic membrane
5. Receptor binding opens sodium ion channels in the postsynaptic membrane, causing depolarisation
6. If sufficient depolarisation occurs (spatial or temporal summation reaches threshold), an action potential is generated in the postsynaptic neurone
7. ACh is rapidly hydrolysed by acetylcholinesterase in the synaptic cleft, stopping the signal
8. Choline is reabsorbed into the presynaptic terminal and recycled to form new ACh

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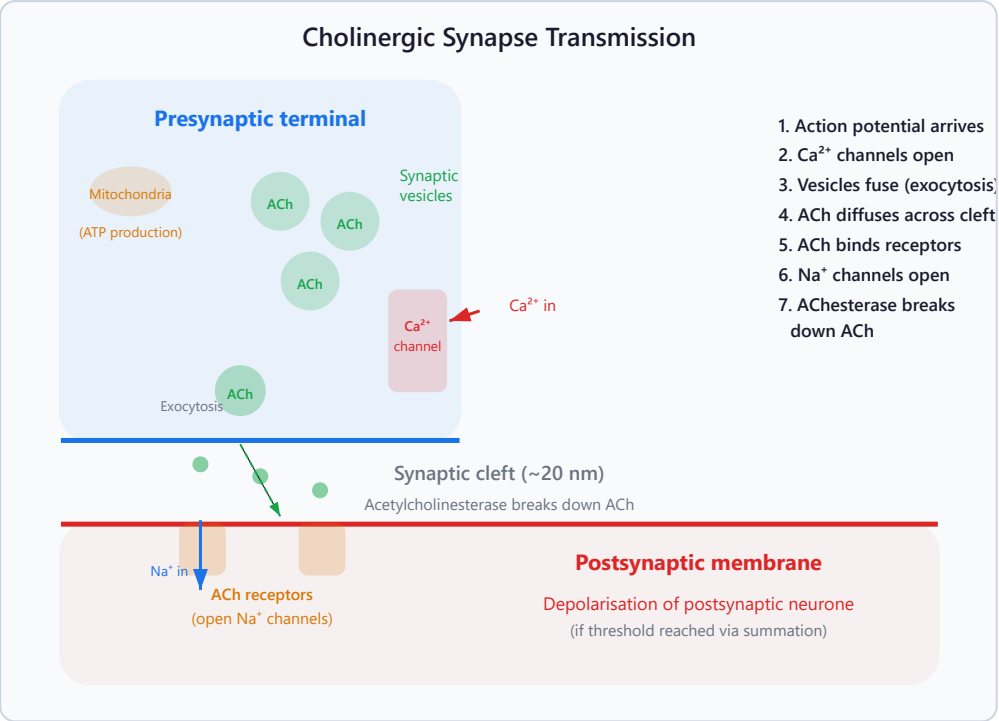

Summation allows weak stimuli to trigger postsynaptic action potentials: temporal summation (repeated impulses from the same presynaptic neurone release enough neurotransmitter to reach threshold) and spatial summation (multiple presynaptic neurones simultaneously release neurotransmitter onto one postsynaptic neurone, collectively reaching threshold).

5.2 Hormonal Communication

The Endocrine System

Hormones are chemical messengers produced by endocrine glands and secreted directly into the blood. They travel throughout the body but only affect target cells that possess specific complementary receptors. Hormonal communication is slower than nervous communication (seconds to hours rather than milliseconds) but produces longer-lasting, widespread effects.

Gland	Hormone	Target/Function
Hypothalamus	Releasing/inhibiting hormones, ADH	Controls anterior pituitary; ADH acts on collecting ducts of kidney

Anterior pituitary	FSH, LH, TSH, ACTH, Growth hormone	Controls other endocrine glands; growth
Posterior pituitary	ADH, Oxytocin (stored here)	Kidney water reabsorption; uterine contractions
Thyroid	Thyroxine (T4)	Metabolic rate regulation
Adrenal medulla	Adrenaline	Fight-or-flight response
Adrenal cortex	Cortisol, Aldosterone	Stress response; sodium reabsorption in kidney
Pancreas (islets of Langerhans)	Insulin (beta cells), Glucagon (alpha cells)	Blood glucose regulation

Hormone Types and Mechanisms

Peptide/protein hormones (e.g., insulin, glucagon, ADH) are water-soluble and cannot cross the cell membrane. They bind to receptors on the cell surface, activating intracellular second messenger systems (e.g., cyclic AMP) that amplify the signal and produce a cellular response through enzyme activation cascades.

Steroid hormones (e.g., oestrogen, testosterone, cortisol) are lipid-soluble and can pass through the cell membrane. They bind to intracellular receptors (often in the cytoplasm or nucleus), forming hormone-receptor complexes that act as transcription factors, directly influencing gene expression by binding to specific DNA sequences.

5.3 Homeostasis

Homeostasis: The maintenance of a constant internal environment within restricted limits despite changes in external conditions. It ensures that cells have optimal conditions for enzyme activity and normal cell function. Homeostatic mechanisms involve receptors (detecting deviations from the set point), a coordination centre (processing information), and effectors (producing corrective responses).

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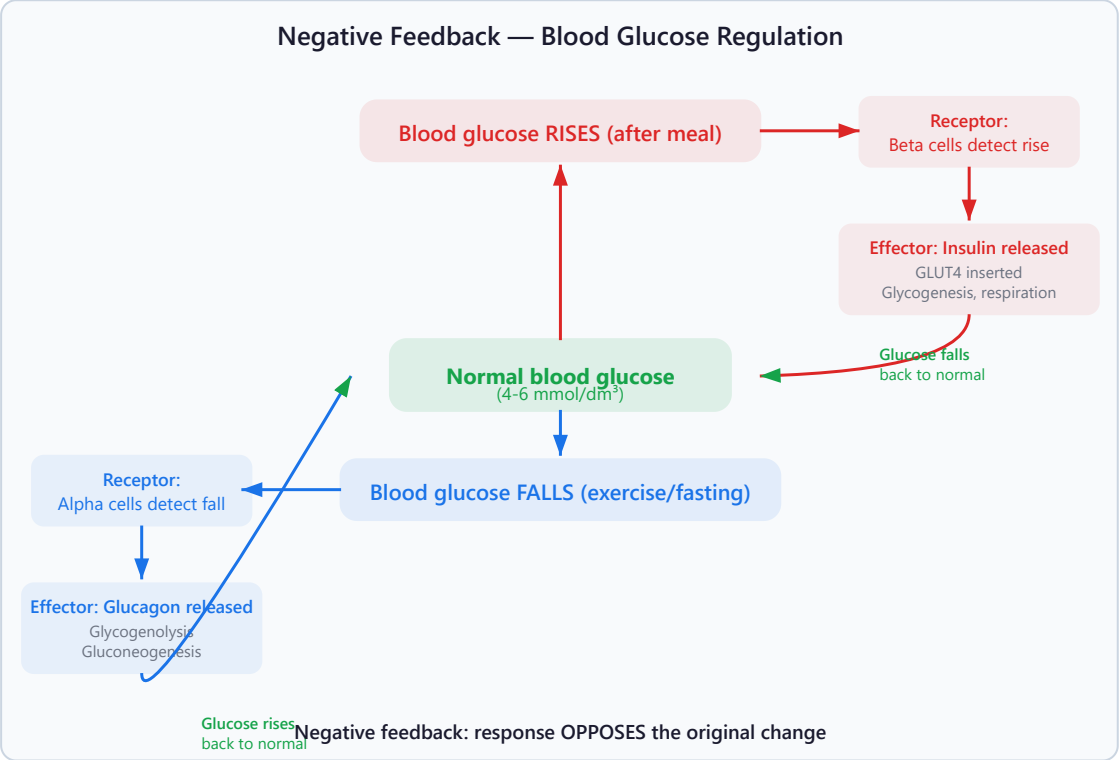
Negative Feedback

Negative feedback is the primary mechanism of homeostasis. When a physiological parameter deviates from its set point (norm), receptors detect the change and effectors are activated to return the parameter to the set point. Crucially, the response opposes the original change: if a parameter rises above the set point, mechanisms act to decrease it, and vice versa. This produces oscillations around the set point rather than a fixed value.



Blood Glucose Regulation

Blood glucose concentration is maintained between approximately 4-6 mmol/dm3 by the antagonistic action of insulin and glucagon, both produced by the pancreatic islets of Langerhans.

When blood glucose rises (after a meal): beta cells detect the increase and secrete insulin. Insulin binds to receptors on liver and muscle cells, stimulating: glucose uptake via insertion of GLUT4 transporters into cell membranes, increased glycogenesis (glucose converted to glycogen for storage), increased cellular respiration of glucose, and increased conversion of glucose to fatty acids. These actions lower blood glucose back towards the set point.

When blood glucose falls (between meals, during exercise): alpha cells detect the decrease and secrete glucagon. Glucagon binds to receptors on liver cells, stimulating: glycogenolysis (glycogen

broken down to glucose), gluconeogenesis (synthesis of new glucose from amino acids and glycerol), and release of glucose into the blood. These actions raise blood glucose back towards the set point.

Diabetes mellitus:

Feature	Type 1	Type 2
Cause	Autoimmune destruction of beta cells	Target cells become resistant to insulin; beta cells may also decline over time
Onset	Usually childhood/adolescence (rapid onset)	Usually adulthood (gradual onset); increasingly seen in younger people
Insulin production	None (or very little)	Initially normal or high, later may decline
Risk factors	Genetic predisposition + environmental trigger (possibly viral)	Obesity (especially visceral fat), physical inactivity, age, family history, ethnicity
Treatment	Insulin injections (or pump); monitoring blood glucose; carbohydrate counting	Lifestyle changes (diet, exercise, weight loss); metformin (increases insulin sensitivity); sulfonylureas (stimulate insulin secretion); eventually insulin may be needed
Prevalence	~10% of diabetes cases	~90% of diabetes cases (growing epidemic worldwide)

Second messenger model of hormone action: Glucagon and adrenaline are water-soluble hormones that cannot cross the cell membrane. They bind to receptors on the cell surface, activating a G-protein which in turn activates the enzyme adenylyl cyclase. This converts ATP to cyclic AMP (cAMP), the second messenger, which activates a cascade of protein kinases inside the cell, amplifying the signal and triggering the cellular response (e.g., glycogenolysis). This explains how a tiny amount of hormone can produce a large intracellular effect.

Thermoregulation

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Core body temperature is maintained at approximately 37 degrees Celsius by the thermoregulatory centre in the hypothalamus, which acts as both the receptor (detecting blood temperature) and the coordination centre.

Response to overheating: vasodilation of skin arterioles (more blood flows near the surface, increasing radiation of heat), sweating (evaporation of water from skin absorbs latent heat), erector pili muscles relax (hairs lie flat, reducing insulating air layer), and reduced metabolic rate.

Response to cooling: vasoconstriction of skin arterioles (less blood near surface, reducing heat loss), shivering (involuntary muscle contractions generate heat from increased respiration), erector pili muscles contract (hairs stand up, trapping insulating air), increased metabolic rate (thyroxine increases cellular respiration), and behavioural responses (adding clothing, seeking warmth).

Osmoregulation and the Kidney

The kidney regulates water potential of the blood through ultrafiltration in the glomerulus (small molecules forced into the Bowman's capsule under high pressure) followed by selective reabsorption in the proximal convoluted tubule (glucose, amino acids, and most water and ions actively and passively reabsorbed) and fine-tuning of water reabsorption in the collecting duct.

The loop of Henle creates a water potential gradient in the medulla through a countercurrent multiplier mechanism. The descending limb is permeable to water (water leaves by osmosis into the hypertonic medulla) but not to ions. The ascending limb is impermeable to water but actively pumps sodium and chloride ions out into the medulla, increasing medullary solute concentration.

ADH (antidiuretic hormone) controls the permeability of the collecting duct to water. When blood water potential is low (dehydration), osmoreceptors in the hypothalamus shrink, stimulating the posterior pituitary to release more ADH. ADH causes aquaporin channels to be inserted into the collecting duct walls, increasing water reabsorption by osmosis into the hypertonic medulla, producing small volumes of concentrated urine. When blood water potential is high, less ADH is released, fewer aquaporins are present, less water is reabsorbed, and large volumes of dilute urine are produced.

> **Exam Tip:** When explaining negative feedback loops, always identify all four components: the stimulus (deviation from set point), the receptor (what detects the change), the effector (what produces the response), and the response itself (which opposes the original stimulus). Then explain how the response returns the parameter to the set point.

5.4 Excretion

The Liver

The liver plays a central role in excretion by deaminating excess amino acids (removing the amino group, which is converted to urea via the ornithine cycle) and detoxifying harmful substances (alcohol, drugs) using cytochrome P450 enzymes in smooth endoplasmic reticulum of hepatocytes. The liver also stores glycogen, produces bile (containing bile salts for fat emulsification and bilirubin from haemoglobin breakdown), synthesises plasma proteins, and regulates blood glucose and lipid levels.

The liver receives a dual blood supply: the hepatic artery delivers oxygenated blood for aerobic respiration of hepatocytes, and the hepatic portal vein delivers nutrient-rich deoxygenated blood directly from the gut for processing. Blood leaves via the hepatic vein. The liver lobule structure with sinusoids ensures all hepatocytes are in close contact with blood for efficient exchange.

5.5 Photosynthesis

Light-Dependent Reactions

The light-dependent reactions occur on the thylakoid membranes of chloroplasts. Light energy is absorbed by photosystems (clusters of chlorophyll and accessory pigments). The process involves:

1. Light energy excites electrons in chlorophyll molecules of Photosystem II (P680), raising them to a higher energy level
2. Excited electrons pass along an electron transport chain (ETC) of carrier proteins, losing energy at each step
3. Energy released by electron flow is used to pump H+ ions from the stroma into the thylakoid lumen, creating a proton gradient
4. H+ ions flow back through ATP synthase by chemiosmosis, driving ATP synthesis (photophosphorylation)
5. Electrons from the ETC arrive at Photosystem I (P700), where they are re-excited by light
6. These high-energy electrons reduce NADP+ (with H+ ions from photolysis) to form reduced NADP (NADPH)

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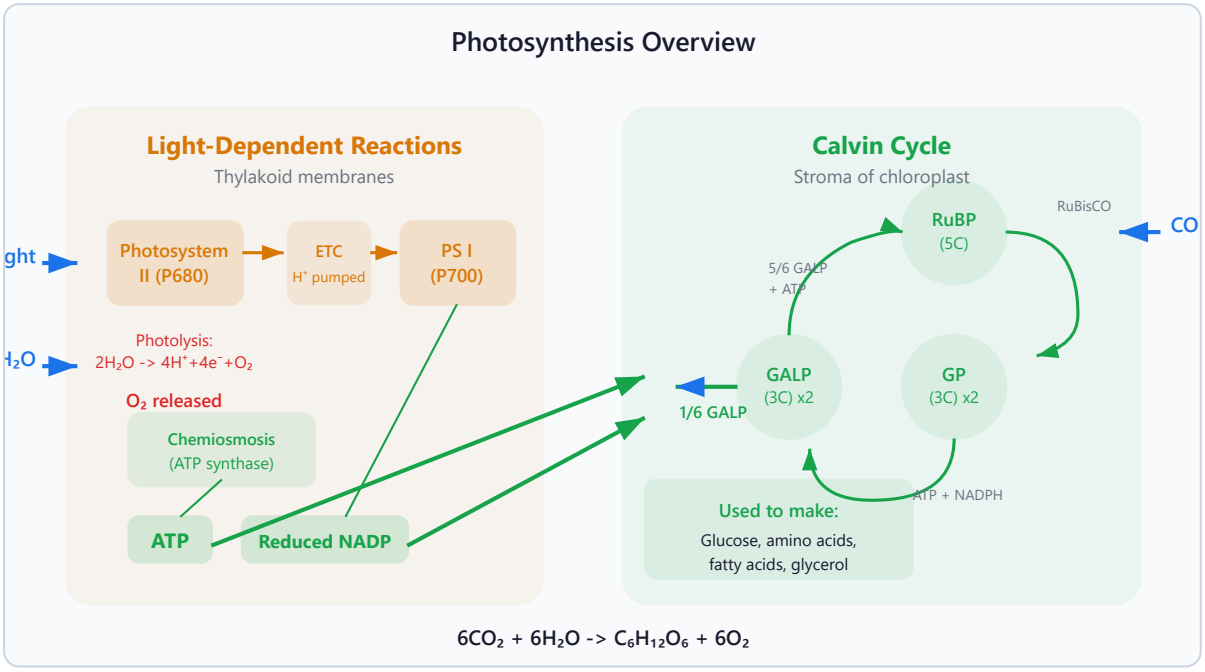
7. Water is photolysed (split by light) to replace electrons lost from PSII: 2H2O produces 4H+ + 4e- + O2

Products of the light-dependent reactions: ATP and reduced NADP (used in the Calvin cycle) and oxygen (waste product released or used in respiration).



Light-Independent Reactions (Calvin Cycle)

The Calvin cycle occurs in the stroma of chloroplasts and does not directly require light, but depends on ATP and reduced NADP from the light-dependent reactions. The stages are:

1. **Carbon fixation:** CO2 combines with ribulose bisphosphate (RuBP, a 5-carbon compound) catalysed by the enzyme RuBisCO, forming an unstable 6-carbon compound that immediately splits into two molecules of glycerate-3-phosphate (GP, 3-carbon)
2. **Reduction:** GP is reduced to glyceraldehyde-3-phosphate (GALP/triose phosphate) using ATP (provides energy) and reduced NADP (provides hydrogen atoms)
3. **Regeneration of RuBP:** Most GALP molecules (5 out of 6) are used to regenerate RuBP using ATP, allowing the cycle to continue. One in six GALP molecules is the net product, used to synthesise glucose, amino acids, fatty acids, and other organic molecules.

Limiting Factors

The rate of photosynthesis is limited by whichever factor is in shortest supply relative to demand (the principle of limiting factors). The main limiting factors are light intensity (affects rate of light-dependent reactions, ATP and NADPH production), carbon dioxide concentration (affects carbon fixation rate in Calvin cycle), and temperature (affects enzyme activity, particularly RuBisCO). In practice, at low light levels, light is limiting; at high light levels with ambient CO2, carbon dioxide becomes limiting; and at high light and high CO2, temperature may become limiting.

5.6 Respiration

Glycolysis

Glycolysis occurs in the cytoplasm and does not require oxygen. One molecule of glucose (6C) is phosphorylated using 2 ATP, then split into two molecules of triose phosphate (3C), which are oxidised to pyruvate (3C). Net yield per glucose: 2 ATP (by substrate-level phosphorylation) and 2 reduced NAD.

The Link Reaction

In aerobic conditions, pyruvate enters the mitochondrial matrix where the link reaction converts it to acetyl CoA (2C) by: oxidative decarboxylation (one carbon removed as CO2, hydrogen atoms transferred to NAD forming reduced NAD), and attachment to coenzyme A. Per glucose: 2 acetyl CoA, 2 CO2, 2 reduced NAD.

The Krebs Cycle

Acetyl CoA (2C) combines with oxaloacetate (4C) to form citrate (6C), which is progressively decarboxylated and dehydrogenated in a cycle of reactions that regenerates oxaloacetate. Per turn of the cycle (per acetyl CoA): 2 CO2 released, 3 reduced NAD produced, 1 reduced FAD produced, 1 ATP produced (by substrate-level phosphorylation). Per glucose (two turns): 4 CO2, 6 reduced NAD, 2 reduced FAD, 2 ATP.

Oxidative Phosphorylation

Oxidative phosphorylation occurs on the inner mitochondrial membrane (cristae) and produces the majority of ATP. Reduced NAD and reduced FAD donate electrons and hydrogen ions to the electron transport chain. Electrons pass along a series of carrier proteins, releasing energy that is used to

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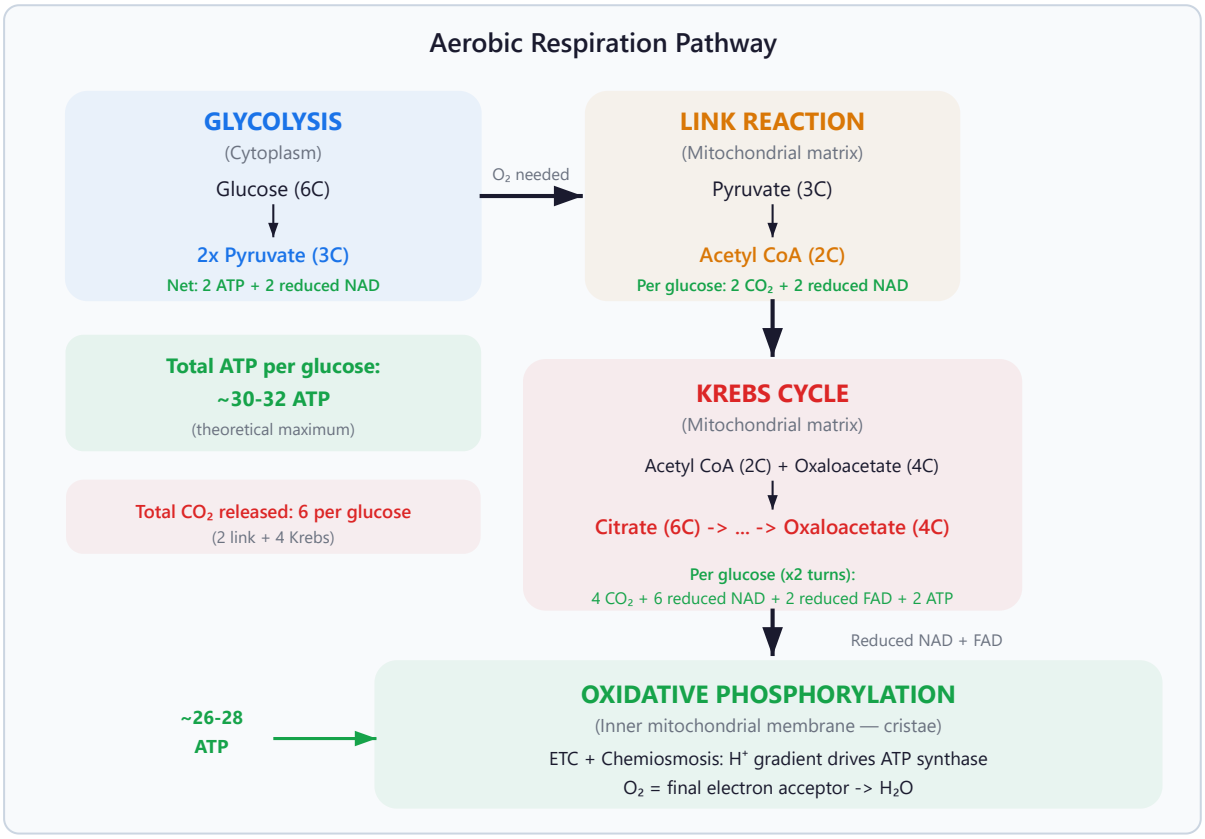
pump H+ ions from the matrix into the intermembrane space, creating a proton motive force. H+ ions flow back through ATP synthase (chemiosmosis), driving the synthesis of ATP from ADP and Pi. Oxygen is the final electron acceptor, combining with electrons and H+ to form water.

Theoretical maximum yield per glucose: 30-32 ATP (actual yield is lower due to proton leakage and ATP used for transport processes). Without oxygen, the electron transport chain stops, reduced NAD and FAD cannot be reoxidised, the Krebs cycle stops, and the cell relies on anaerobic respiration (fermentation) for ATP.



Anaerobic Respiration

In the absence of oxygen, NAD must be regenerated for glycolysis to continue. In animals, pyruvate is reduced to lactate by lactate dehydrogenase, oxidising reduced NAD back to NAD. In plants and yeast, pyruvate is decarboxylated to ethanal (acetaldehyde) then reduced to ethanol, regenerating NAD. Both pathways yield only 2 ATP per glucose (from glycolysis alone), making them far less efficient than aerobic respiration.

> **Respiratory Quotient (RQ):** The ratio of CO2 produced to O2 consumed during respiration. RQ = CO2 produced / O2 consumed. For glucose (purely aerobic): RQ = 6/6 = 1.0. For fatty

acids: RQ is approximately 0.7 (more oxygen needed relative to CO_2 produced because fats are more reduced). For proteins: RQ is approximately 0.9. An RQ greater than 1.0 indicates anaerobic respiration is occurring alongside aerobic.

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